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PaddockTS: paddock-level satellite time series analysis of agroecosystem dynamics

John T. Burley [1¶], **Yasar Adeel Ansari**[1], **Christopher Bradley** [1], **Alex Norton** [1,2], **and Justin Borevitz** [1]

1 Research School of Biology, Australian National University, 46 Sullivans Creek Rd, Acton, ACT 2601, Australia **2** CSIRO Environment, Aspendale, VIC, Australia ¶ Corresponding author

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Software

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Summary

PaddockTimeSeries (PaddockTS) is an open-source data analysis pipeline that aligns Earth observation data with agricultural fields, or paddocks—the core land management units where agricultural decisions are made and outcomes such as yield, phenology, productivity, and soil condition are recorded. PaddockTS enables scalable agricultural monitoring and research on how land management and environmental variation shape agroecosystem dynamics. For a user-defined location and date range in Australia, PaddockTS delineates paddock boundaries from Sentinel-2 time-series imagery using geospatial segmentation or accepts user-provided boundaries, aggregates Sentinel-2 satellite time series within them, and returns analysis-ready datasets and graphics describing vegetation dynamics and derived seasonal features. It also retrieves environmental covariates, including terrain, soil, weather, and water-balance variables. By summarising vegetation dynamics at the paddock level, PaddockTS makes Earth observation data easier to integrate with management records, environmental conditions, and agricultural and ecological outcomes. Its outputs enable comparisons across paddocks, years, and environmental gradients and provide analysis-ready inputs for agroecosystem modelling and machine-learning prediction tasks. PaddockTS can be used as a programmable Python workflow for reproducible research or through a web interface for exploratory analysis without coding.

Software repository: https://github.com/johnburley3000/paddocktimeseries
Interactive web tool: https://www.paddocktimeseries.net/

Statement of need

Satellite data are increasingly used to map crops, detect agricultural management and plant phenology, and estimate field-level productivity and yield (Burke & Lobell, 2017; d'Andrimont et al., 2020; Van Tricht et al., 2023; Xie et al., 2024). Their repeated, globally consistent observations create opportunities to compare how agroecosystems respond to management and environmental variation across large areas, complementing the scope of purpose-built agronomic experiments (Cambron et al., 2024; Hu et al., 2024; Weiss et al., 2020). Such comparisons are important for evaluating how agricultural management in different contexts affects productivity, resilience to climate extremes, and carbon stocks (Guan et al., 2023; Lobell et al., 2025; Nouri et al., 2021; Zhou et al., 2022).

Realising this potential requires satellite observations to be organised at spatial scales appropriate to research questions or applications. Remote-sensing analyses commonly operate at the pixel level or more broadly across land-use categories or administrative regions. However, practices such as sowing, grazing, fertilisation, irrigation, and harvest—and associated records of crop development, soil properties, and yield—are often organised by paddock. Paddock-level

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summaries therefore provide a complementary scale of analysis that aligns satellite time series with the spatial units in which management is applied and outcomes are commonly recorded. Converting dense satellite archives into standardised paddock-level time series also makes comparisons more tractable at continental or global scales.

PaddockTS addresses this need by providing a reproducible workflow for generating paddock-level time series and derived features together with relevant environmental covariates. It is intended for researchers conducting agroecosystem monitoring, comparative analysis, machine learning, and process-based modelling, as well as practitioners seeking interpretable summaries of satellite and environmental data that complement local knowledge. The current implementation focuses on Australia because it builds on Australian Earth observation and environmental datasets, while its modular architecture supports extension to other regions and data sources.

State of the field

PaddockTS builds on several open-access Earth observation datasets and analytical tools. It retrieves Sentinel-2 NBART surface reflectance through Digital Earth Australia (Dhu et al., 2017; Geoscience Australia, 2022) and environmental covariates from Copernicus DEM, SILO, SLGA, and OzWALD (Dijk et al., 2018; European Space Agency, 2022; Jeffrey et al., 2001; Viscarra Rossel et al., 2015). It also applies established spectral indices, an Australian fractional-cover model (Scarth et al., 2010, 2022), and established approaches for estimating phenology metrics (Jönsson & Eklundh, 2004; Trotter & Robinson, 2021).

Paddock segmentation presents a distinct challenge within this workflow and is an active area of research in satellite remote sensing (Corley et al., 2026; Muhawenayo et al., 2026). Recent approaches increasingly use time-series satellite data to exploit differences in vegetation trajectories and management timing (Watkins & Niekerk, 2019; Yan et al., 2024), although performance remains dependent on agricultural context, imagery, and model parameterisation (Ferreira et al., 2025). Because paddocks are transient in time and are not always clearly defined spatial units, particularly in heterogeneous cropping–pasture landscapes, this motivates flexible workflows that support different automated segmentation approaches as well as user-supplied polygons to define paddocks.

PaddockTS's contribution is to integrate otherwise separate capabilities into a scalable workflow for time-series analysis of agroecosystem dynamics, with paddocks as the core spatial unit of analysis. It complements increasingly available field-boundary predictions (e.g., Robinson et al., 2026) and benchmarking datasets (e.g., Kerner et al., 2024). It differs from platforms that support data access and custom processing but do not provide this domain-specific workflow and its standardised outputs by default (Gorelick et al., 2017; Haan et al., 2023).

Software design

Workflow architecture

PaddockTS is organised as a modular workflow that coordinates data acquisition, paddock delineation, time-series construction, seasonal feature extraction, and visualisation (Figure 1). Each analysis is defined by an immutable query object, called `troi`, specifying the time and region of interest. Maintained in the separate `troi` package, this object provides a consistent identity for data retrieval and caching across related geospatial packages.

Workflows begin from either a user-defined region with automated paddock delineation or user-supplied paddock polygons. Processing steps can be run as a complete pipeline or independently, with cached intermediate products allowing analyses to resume without recomputation.

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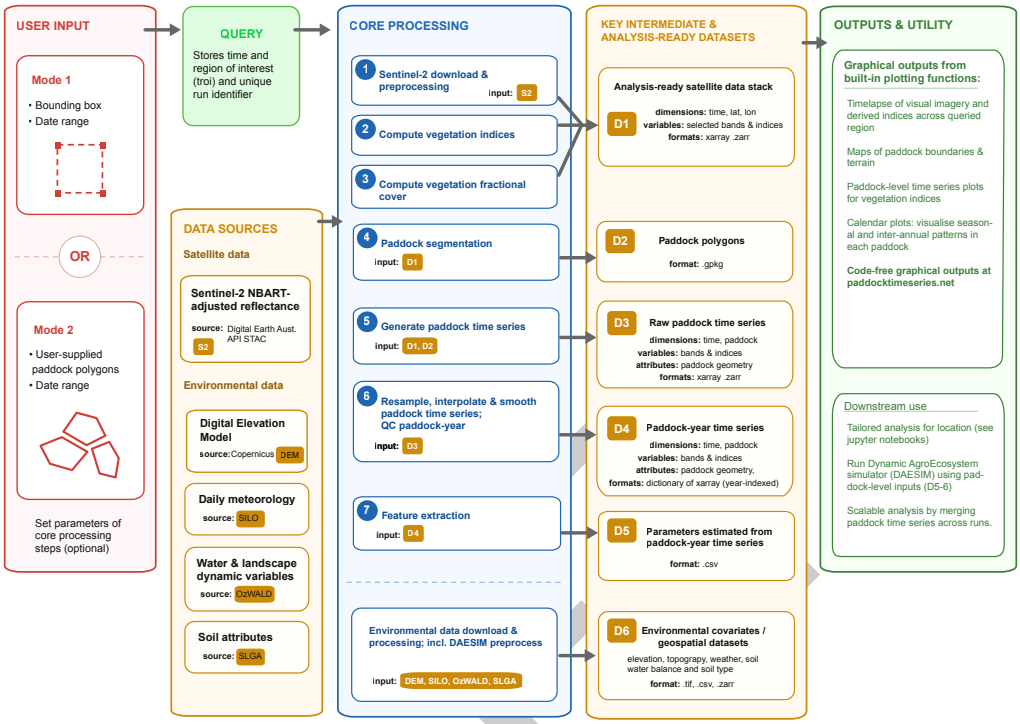


Figure 1: PaddockTS architecture and typical workflows.

Earth observation and environmental data processing and caching

Sentinel-2 NBART surface-reflectance data are accessed from Digital Earth Australia through the independently maintained `pysentinel2` package. Raw reflectance bands and Fmask classifications are stored in a sparse, spatially indexed Zarr data cube. For each request, only missing date/location chunks are downloaded. Overlapping or adjacent queries can therefore reuse locally available observations. Cloud and shadow masking uses a configurable procedure within `pysentinel2`, allowing independent testing and development. Cleaned reflectance data are then used to calculate spectral indices and vegetation fractional cover, which are also cached for downstream analyses.

Terrain, soil, weather, and water-balance data are retrieved through dedicated packages for Copernicus DEM, SLGA, SILO, and OzWALD. These packages apply source-appropriate spatial and temporal caching and return products aligned to the query region and period. Because some environmental datasets are substantially coarser than Sentinel-2 imagery, they are treated as contextual or modelling covariates rather than assumed to represent within-paddock variation.

Paddock segmentation

The default workflow applies a Fourier transform to the normalised difference water index (NDWI) time series to produce a three-band image summarising temporal variation, which is supplied to SAMGeo for segmentation (Kirillov et al., 2023; Wu & Osco, 2023). As in other time-series-based approaches, this representation identifies spatial units with relatively consistent temporal dynamics that differ from their immediate surroundings. Candidate paddock boundaries may vary with the query date range, reflecting both genuine changes in land management and uncertainty in the segmentation, including omission and commission errors. PaddockTS therefore treats automated segmentation as a pragmatic means of approximating field boundaries at scale rather than as a definitive representation of management units. Predicted polygons can be filtered geometrically and by vegetation-index trajectories to

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reduce commission errors. Applications requiring reliable boundary accuracy should evaluate segmentation results against appropriate reference data or substitute user-provided boundaries.

Paddock time series and seasonal features

Paddock polygons are aligned to the Sentinel-2 grid to aggregate median reflectance, spectral-index, and fractional-cover values by paddock and acquisition date. The resulting series can be resampled, interpolated, smoothed, and quality checked using configurable settings, while both observed and processed trajectories are retained. Processed time series are also divided into paddock-year datasets containing calendar-date and day-of-year coordinates. Processed NDVI trajectories for each paddock-year are passed to PhenoloPy to estimate the start, peak, and end of the growing season.

Data products and visual outputs

PaddockTS preserves outputs from multiple processing stages, including satellite reflectance, paddock polygons, paddock-by-time and paddock-year datasets, seasonal features, and environmental covariates. These analysis-ready data products are used by PaddockTS to generate maps, animations of satellite imagery, annotated paddock calendars, landscape-scale time-series summaries, environmental diagnostics, and phenology plots. Persistent paddock identifiers allow time series and derived features to be joined with user-provided attributes and observations, such as crop type, management activities, yield, or soil measurements.

Demonstration

The workflow was applied to a mixed cropping–grazing landscape near Milgadara, New South Wales, Australia, where paddock boundaries and management histories were available. Figure 2 compares user-provided paddock boundaries to those detected by PaddockTS using 2018–2025 Sentinel data. Paddock-level time series of vegetation fractional cover reveal contrasting trajectories among paddocks and years, which can be viewed alongside weather and soil-moisture conditions (Figure 3). At the individual-paddock scale, calendar plots arrange satellite observations by week and year, allowing vegetation development and management transitions to be interpreted alongside user-provided crop labels (Figure 4). Paddock-year vegetation-index trajectories can also be converted into seasonal features, including the estimated start, peak, and end of the growing season (Figure 5). Together, these outputs demonstrate how PaddockTS supports movement between landscape-scale comparison, detailed inspection of individual paddocks, and analysis-ready seasonal features.

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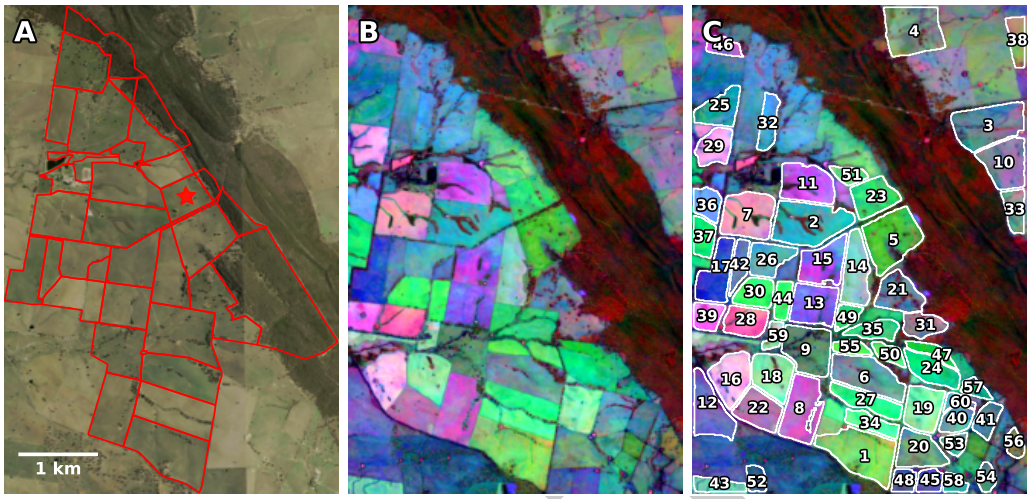


Figure 2: Paddock boundaries for the PaddockTS demonstration landscape near Milgadara, New South Wales, Australia. (A) User-provided paddock boundaries overlaid on aerial imagery. (B) False-colour representation of the Sentinel-2 NDWI time-series Fourier transform (2018–2025) used for segmentation; (C) resulting candidate paddock boundaries numbered from largest to smallest.

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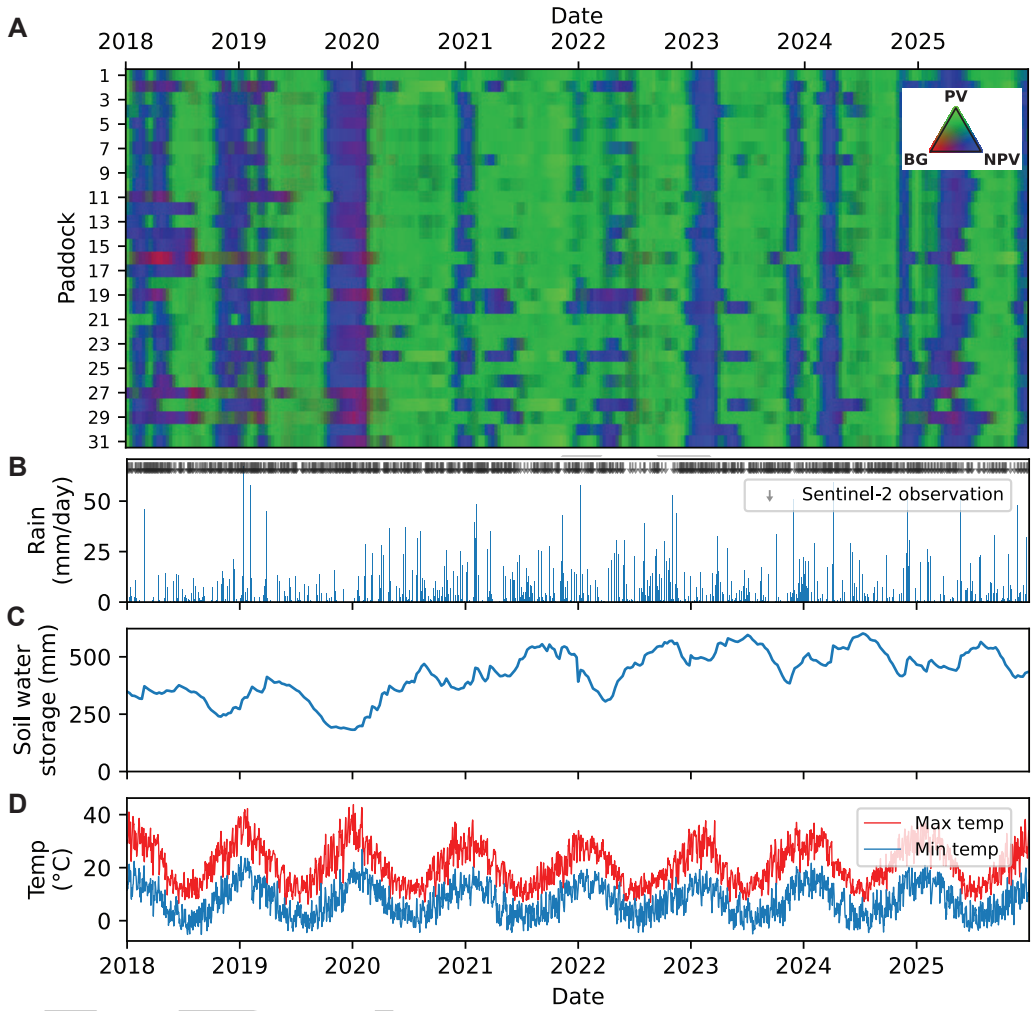


Figure 3: (A) Paddock-level temporal variation in vegetation fractional cover for the 31 largest paddocks identified in the demonstration run. The colour scale represents the predicted fractional contributions of bare ground (BG; red), photosynthetic vegetation (PV; green), and non-photosynthetic vegetation (NPV; blue). Paddock identifiers correspond to Figure 2C. (B–D) Contextual rainfall, soil-moisture and temperature data, and timing of Sentinel-2 observations.

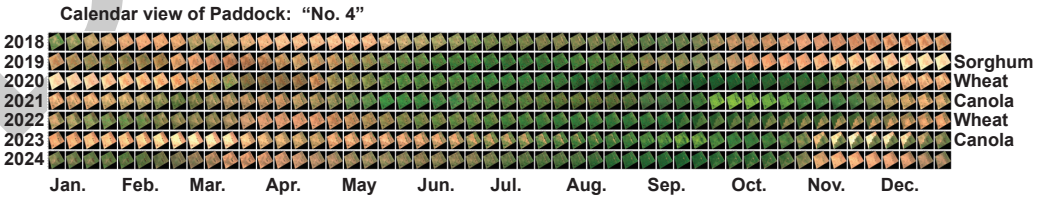


Figure 4: "Calendar view" of an example paddock from the demonstration with recorded crop rotations, marked by a red star in Figure 2A. Sentinel-2 thumbnail images show RGB derived from the temporally resampled, interpolated, and smoothed series. Seasonal variation in plant development and senescence is evident, such as canola flowering visible in 2021 and 2023.

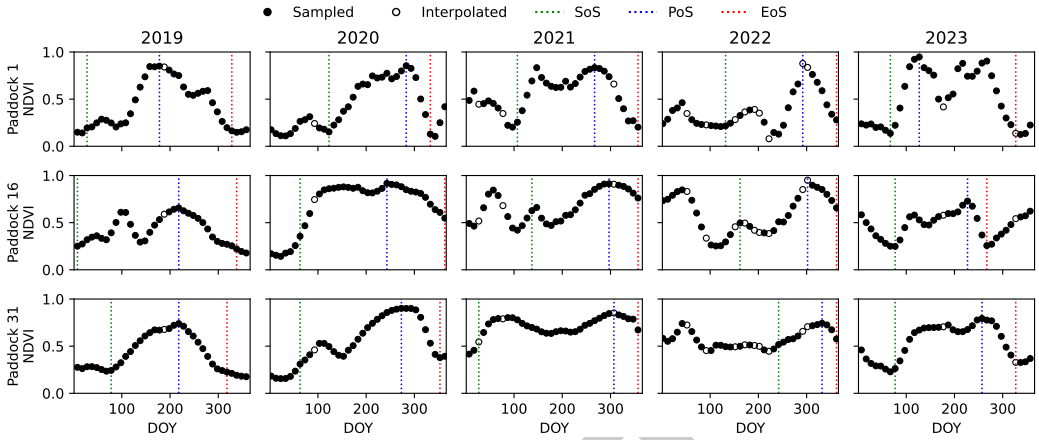


Figure 5: Examples of seasonal features estimated from paddock-level vegetation trajectories. Panels show processed NDVI time series for selected paddock-years at Milgadara, with start of season (SoS), peak of season (PoS), and end of season (EoS) indicated.

Research impact statement

PaddockTS facilitates large-scale analysis of vegetation dynamics and their relationships with agricultural management and environmental variation. Using automated paddock segmentation, PaddockTS has been applied to generate paddock-year datasets across Australia (Bradley, ongoing work), with ongoing efforts to develop derived products for broader research use. Standardised paddock-level time series provide inputs for machine-learning predictions of crop type, yield, and vegetation phenology, while PaddockTS also generates visualisations to support data interpretation and labelling. Paddock-level time series are also being integrated with the DAESim process-based agroecosystem model (Norton & Borevitz, 2025, 2026; Taghikhah et al., 2022), enabling comparisons between observed and simulated vegetation dynamics and supporting future investigations of management and climate scenarios. PaddockTS is also being combined with the Shelterbelts software (Bradley et al., 2026) to investigate effects of farmland trees on agricultural productivity at the paddock scale.

At smaller spatial scales, PaddockTS has been applied with Australian land-management organisations using user-supplied polygons to define management units. At Milgadara, satellite-derived vegetation dynamics were analysed alongside paddock management histories, with the results featured in a public case study by Soils for Life. In collaboration with the Mulloon Institute, PaddockTS has been used to generate remote-sensing data for a catchment-scale study of landscape rehydration, supporting comparisons between treatment and reference areas. These applications demonstrate how PaddockTS supports applied research on land-management interventions and communicates satellite-based insights to practitioners.

AI usage disclosure

Generative AI tools (ChatGPT, OpenAI; Claude, Anthropic) were used to assist in developing PaddockTS code and this manuscript. All AI-assisted outputs were critically reviewed, edited, tested, and validated by the authors. The authors made all scientific, conceptual, and software-design decisions and take full responsibility for the accuracy and integrity of the submitted work.

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